

Vivisecting Starlink Throughput: Measurement and Prediction



Zikun Liu



**Fan-Xue Gabriella
Reidys**



Sarah Tanveer*



Deepak Vasisht



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN



THE UNIVERSITY
of
WISCONSIN
MADISON

LEO constellations are rapidly growing



Starlink is the largest constellation by far

- 6000+ Satellites
- ~2.7M Users
- 42,000 additional incoming

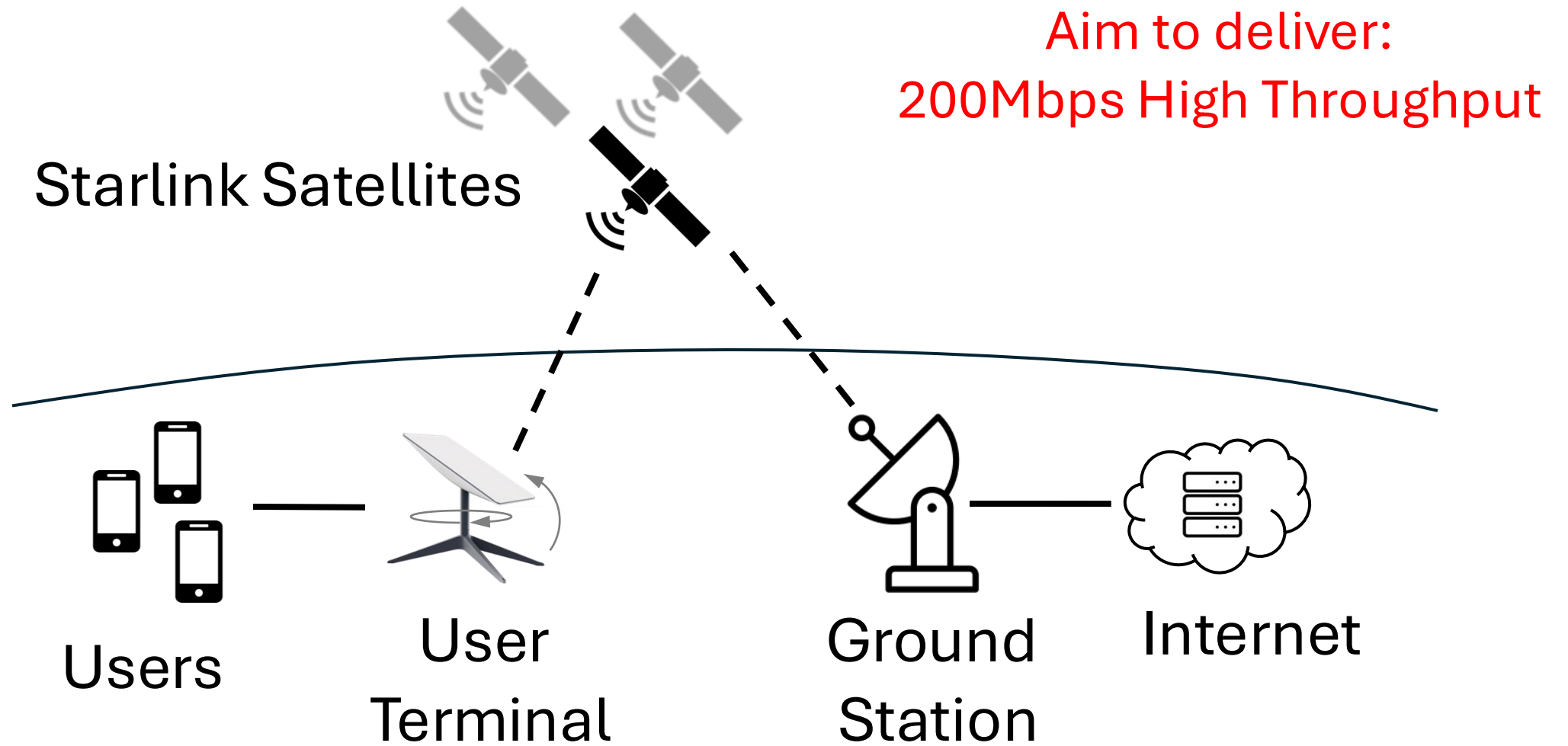
Understanding Starlink throughput

- What factors impact user-experienced throughput?
- Can we predict the throughput in satellite networks like Starlink?

Understanding Starlink throughput

- What factors impact user-experienced throughput?
- Can we predict the throughput in satellite networks like Starlink?

Starlink overview



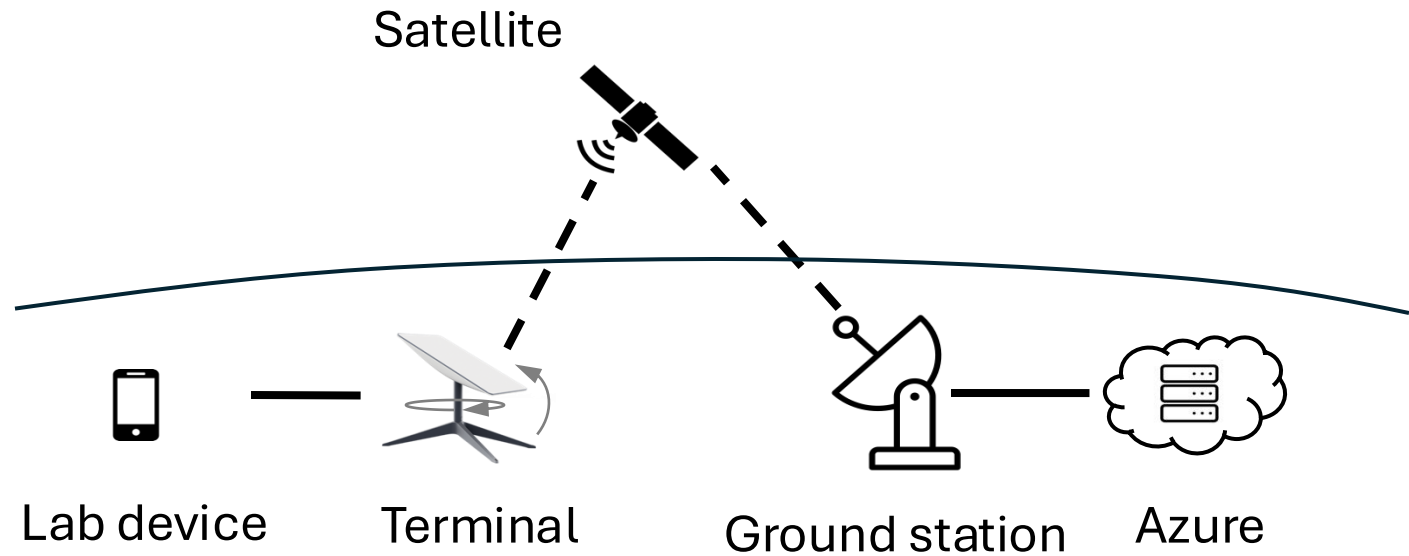
Measuring Starlink's throughput

Across 3 Countries for 6 months

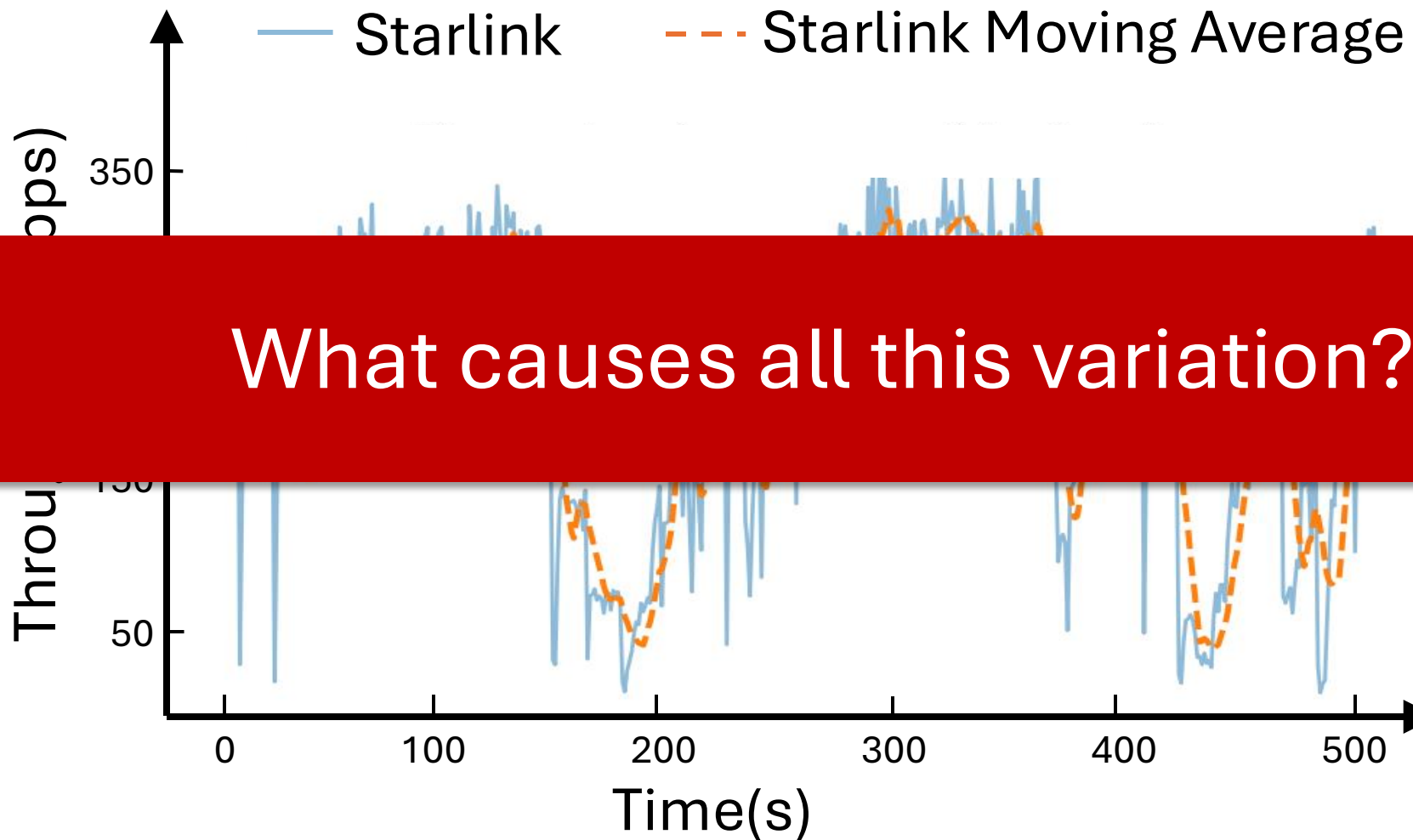
	USA	Canada	Germany
Dishy Version	rev3_proto2	rev3_proto2	rev3_proto2
# of unique Serving Satellites	6,052	3,166	3,956
# of Satellite Handovers	86,808	7,257	26,782
Duration	6 months	1 month	1 month
Cumulative Throughput Traces	41,252 min	2,417 min	10,221 min
Total Throughput Samples	2,475,163	145,053	613,295

Measuring Starlink's throughput

Satellite Network



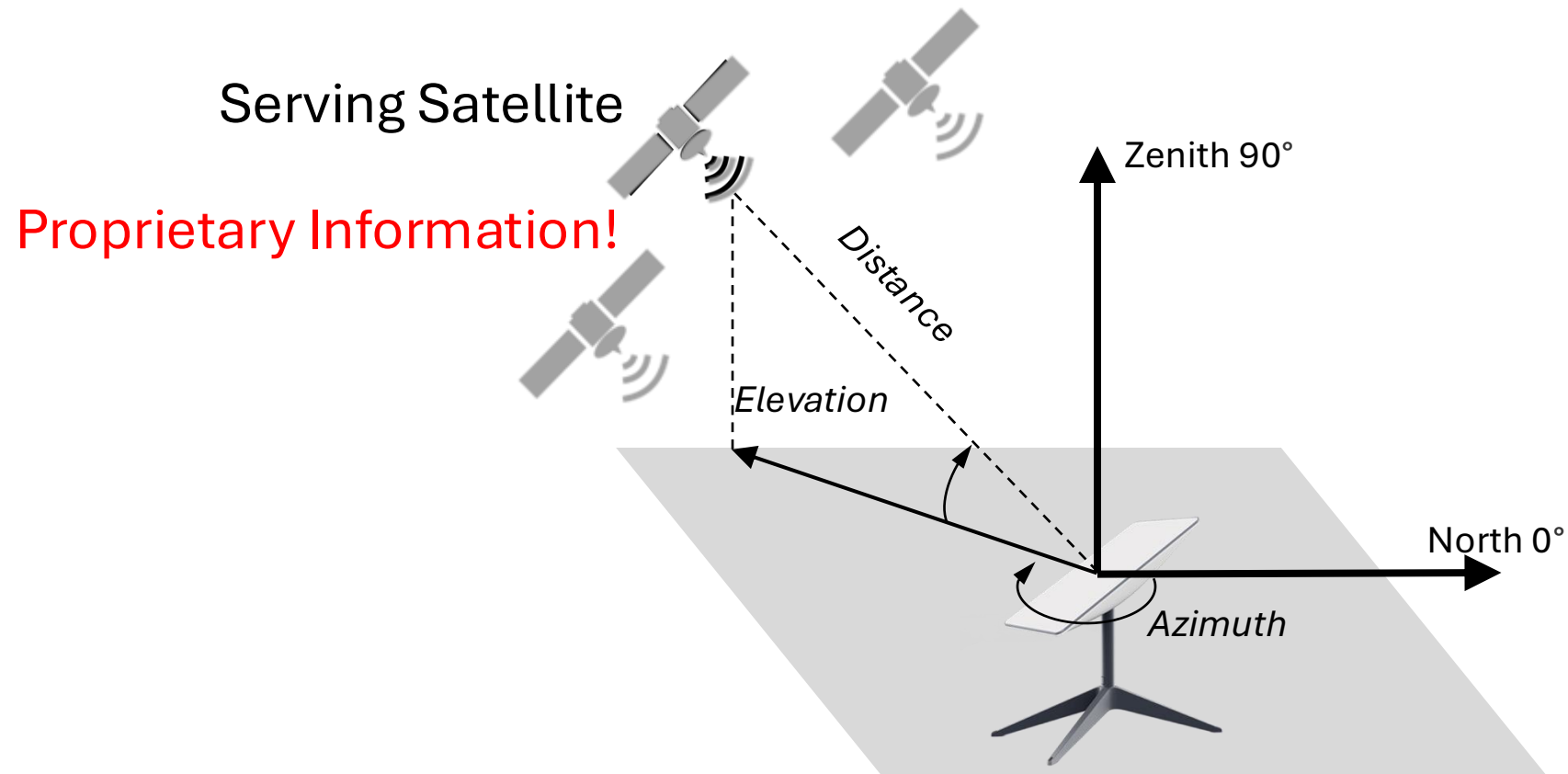
Starlink throughput shows large volatility



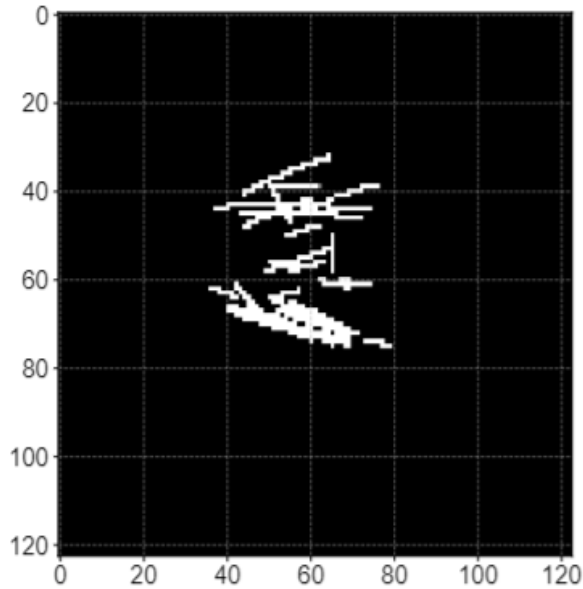
Understanding Starlink throughput

- What factors impact user-experienced throughput?
 - Serving satellite?
- Can we predict the throughput in satellite networks like Starlink?

Challenge: Starlink does not reveal serving satellite

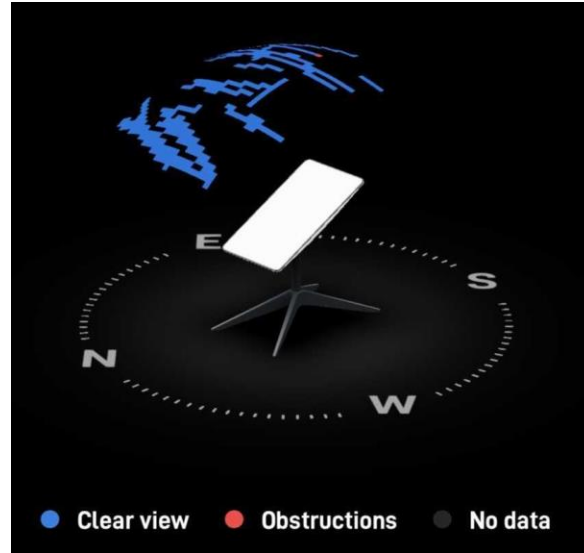


Solution: Infer serving satellite through 2D-3D mapping

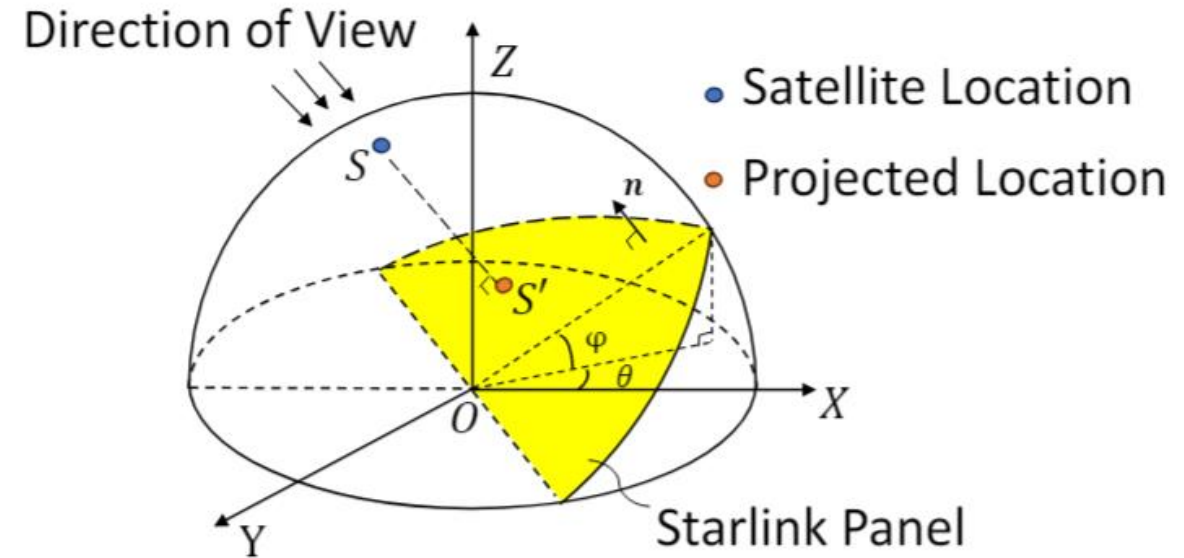


2D Map

X-Y coordinates
hard to interpret!



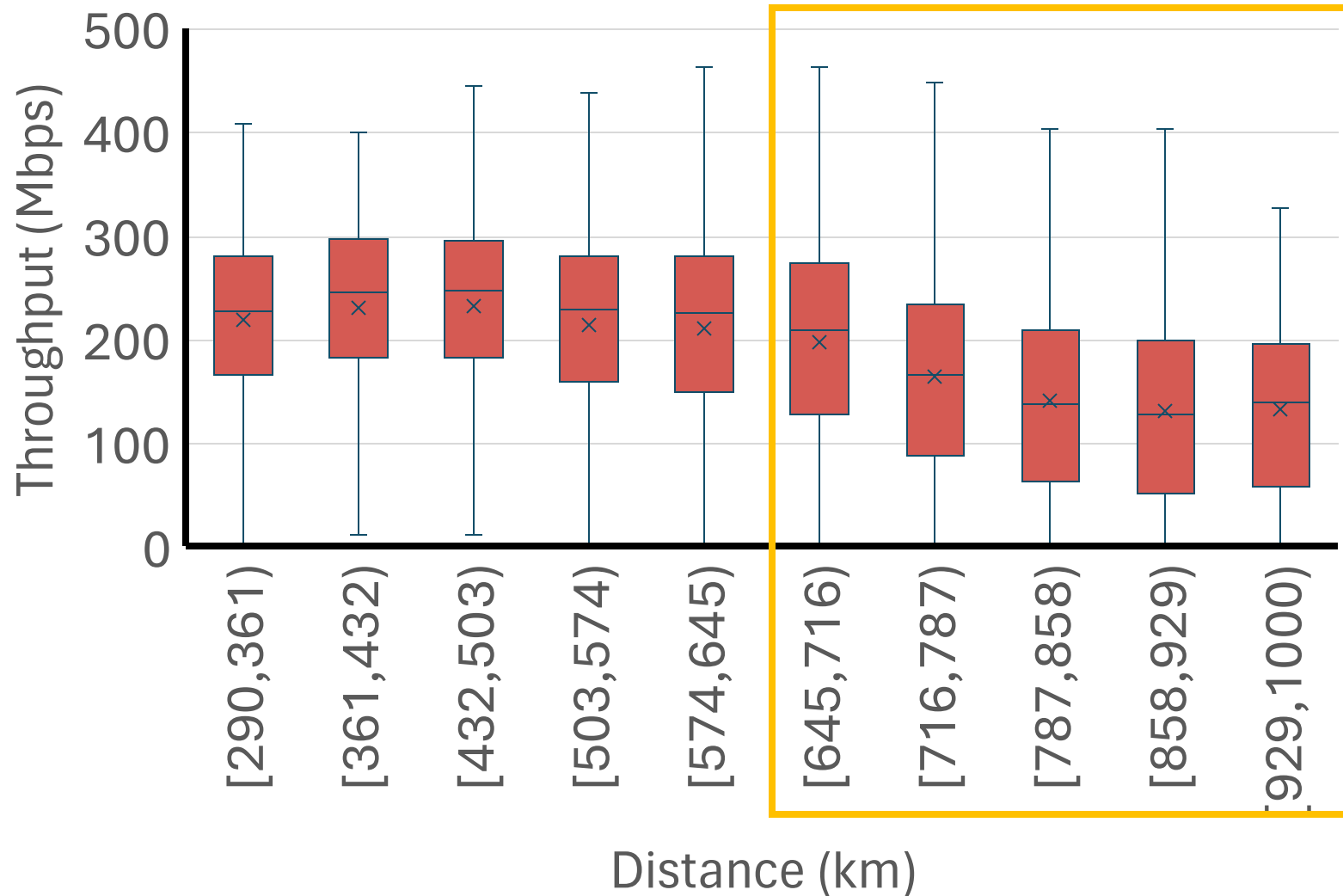
3D Map



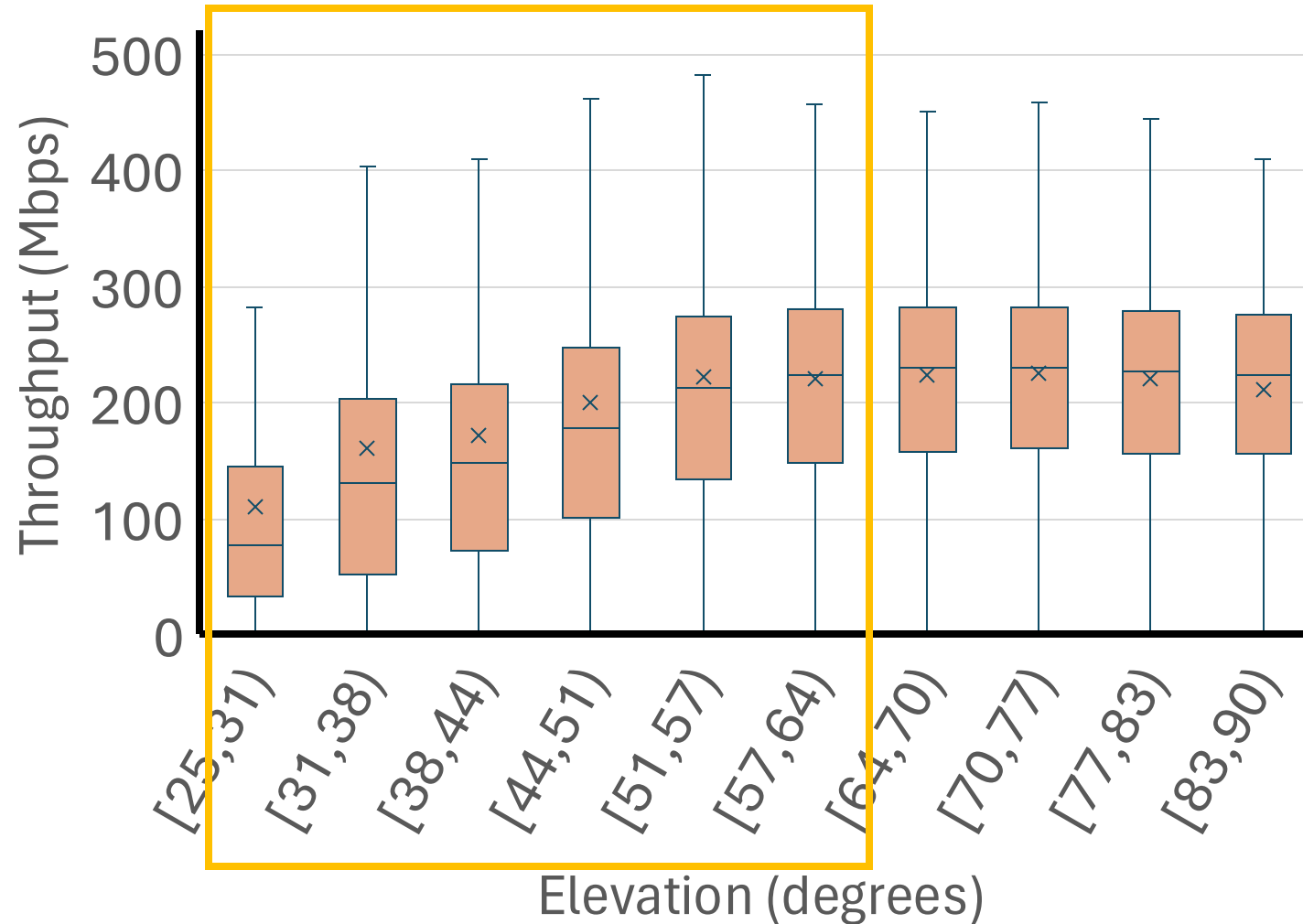
2D – 3D Mapping

Reconstruct 3D
satellite positions!

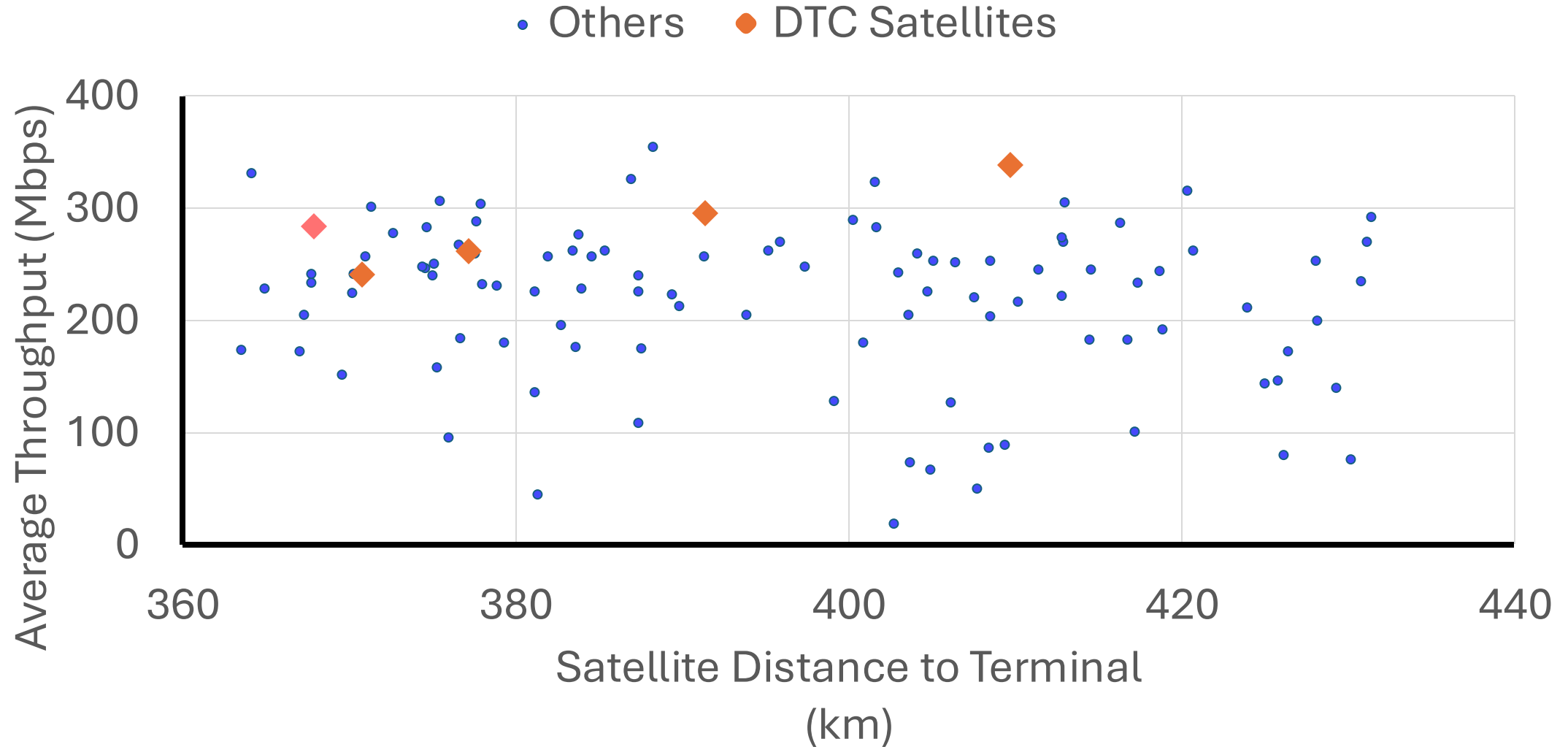
Distance to serving satellite reduces throughput



Higher elevation angle leads to higher throughput



Newer satellite generation offers higher throughput



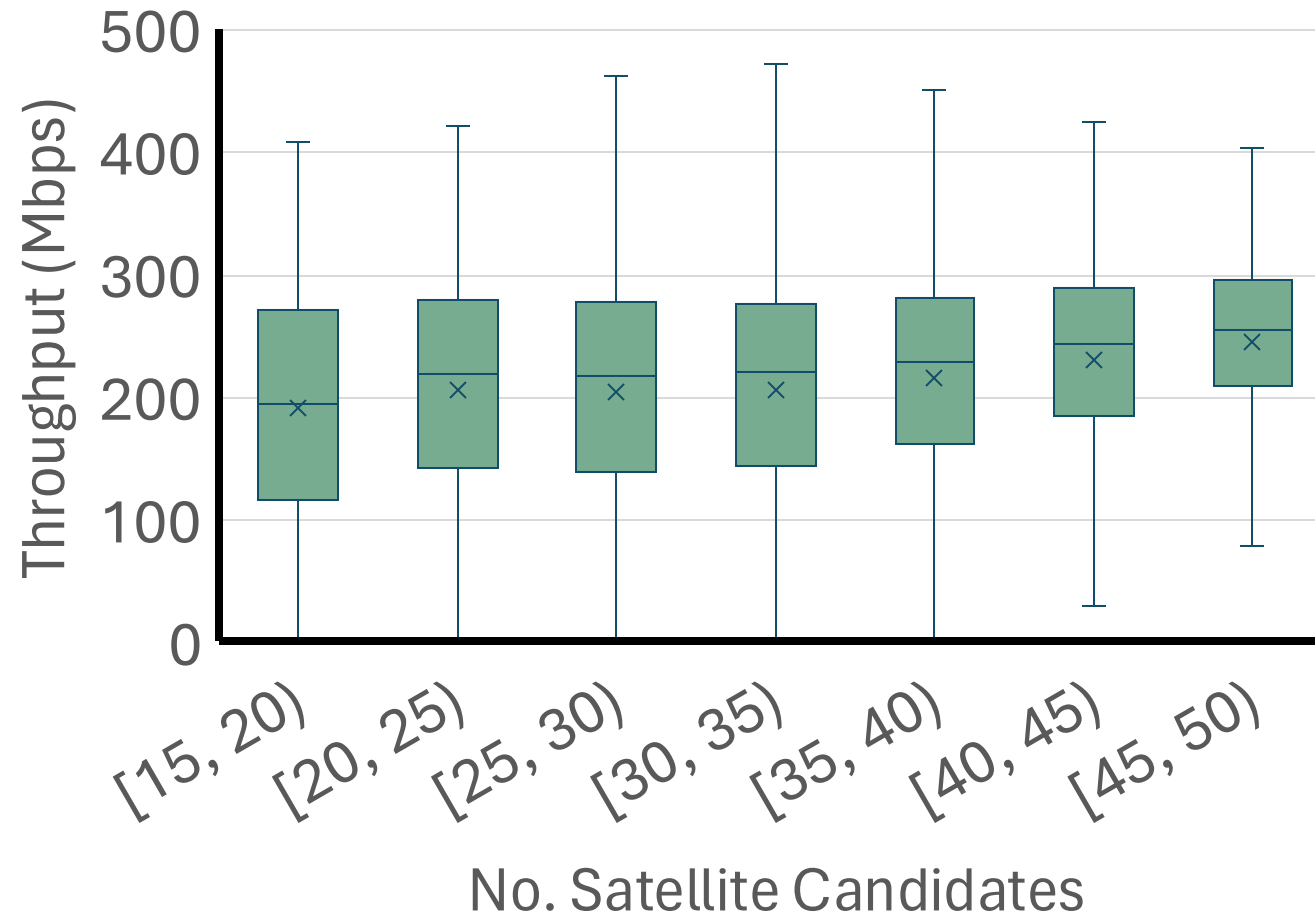
Understanding Starlink throughput

- What factors impact user-experienced throughput?
 - Serving satellite: distance, elevation, and generation matter
- Can we predict the throughput in satellite networks like Starlink?

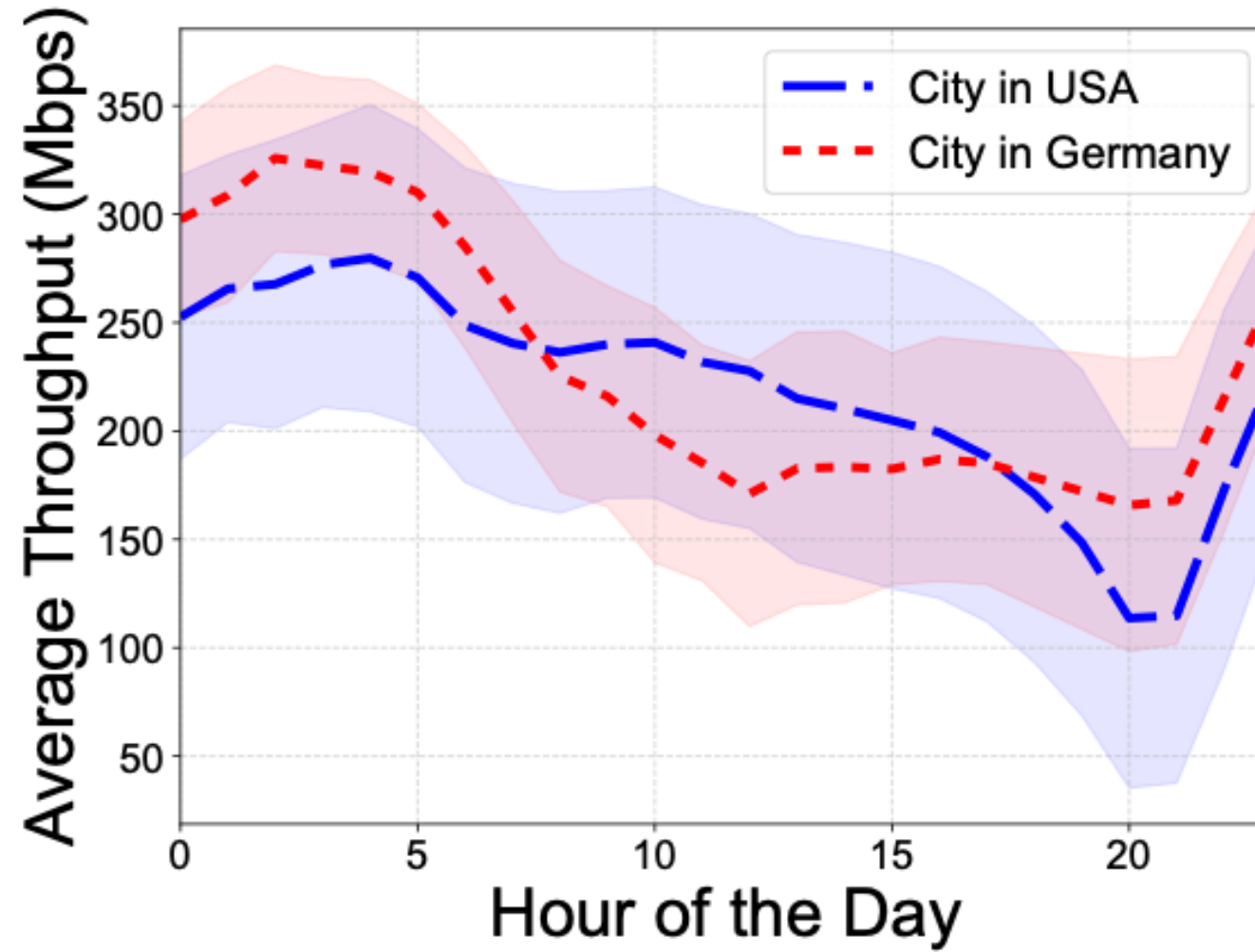
Understanding Starlink throughput

- What factors impact user-experienced throughput?
 - Serving satellite: distance, elevation, and generation matter
 - Load balancing?
- Can we predict the throughput in satellite networks like Starlink?

More available satellites → More throughput



More load → Less throughput



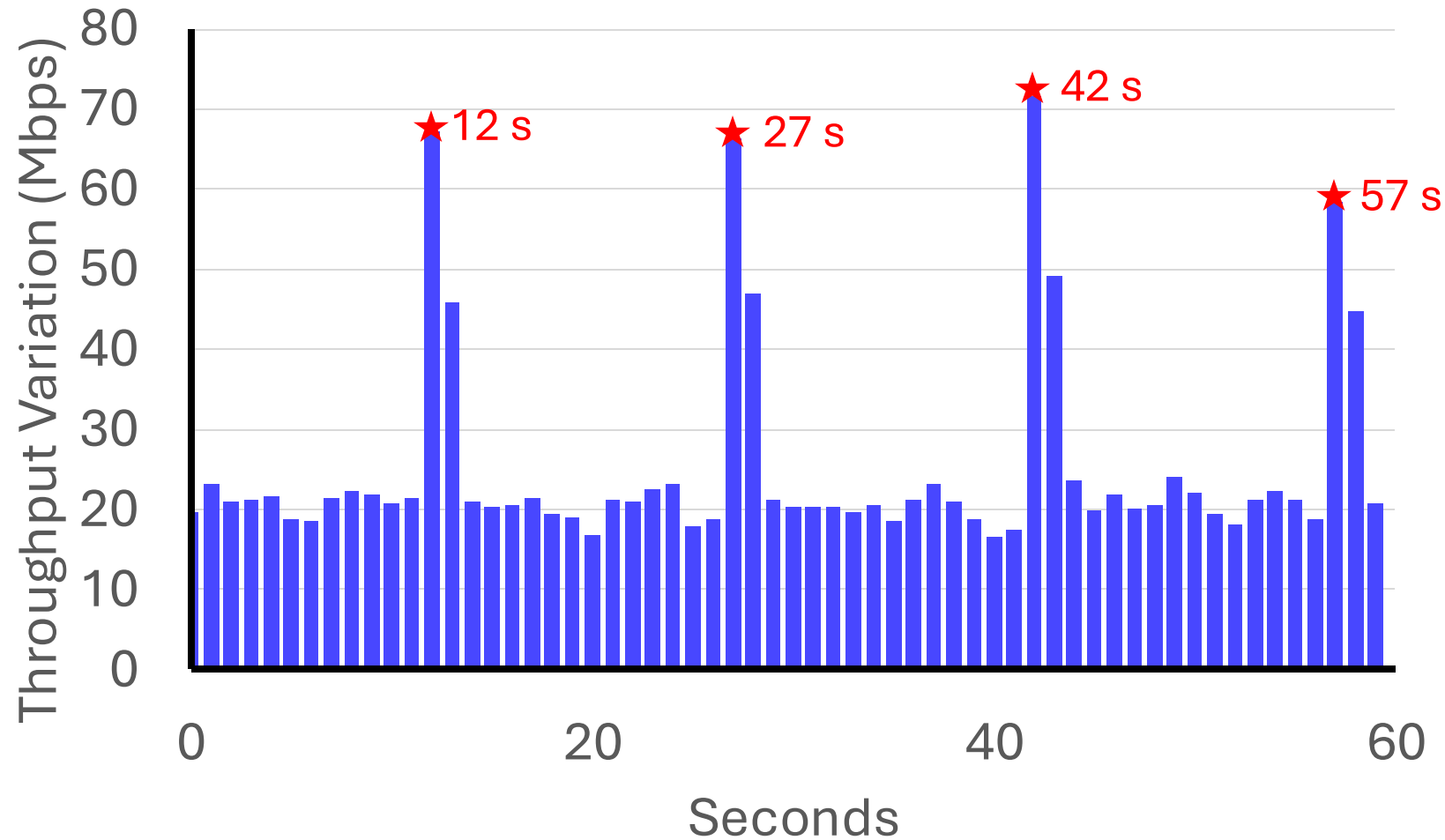
Understanding Starlink throughput

- What factors impact user-experienced throughput?
 - Serving satellite: distance, elevation, and generation matter
 - Load balancing: no. candidate satellites, and hour of the day
 - Weather: rains and clouds
- Can we predict the throughput in satellite networks like Starlink?

Understanding Starlink throughput

- What factors impact user-experienced throughput?
 - Serving satellite: distance, elevation, and generation matter
 - Load balancing: no. candidate satellites, and hour of the day
 - Weather: rains and clouds
 - Global scheduler?
- Can we predict the throughput in satellite networks like Starlink?

Impact of Global Scheduler



Starlink Throughput Shows Periodicity

Understanding Starlink throughput

- What factors impact user-experienced throughput?
 - Serving satellite: distance, elevation, and generation matter
 - Load balancing: no. candidate satellites, and hour of the day
 - Weather: rains and clouds
 - Global scheduler: periodic pattern
- Can we predict the throughput in satellite networks like Starlink?

Understanding Starlink throughput

- What factors impact user-experienced throughput?
 - Serving satellite: distance, elevation, and generation matter
 - Load balancing: no. candidate satellites, and hour of the day
 - Weather: rains and clouds
 - Global scheduler: periodic pattern
- Can we predict the throughput in satellite networks like Starlink?

Throughput prediction improves network QoE

With prediction error $\leq 20\%$, video streaming QoE can be improved close to optimal ($>85\%$)*

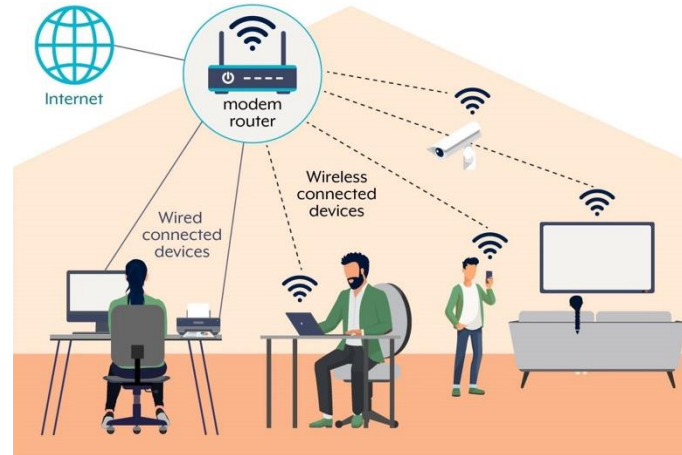


Throughput Prediction is well-known in other networks



LTE/5G

Lumos5G, TMC'22, ...



Wi-Fi

IEEE Access'21, WNJ'19,...

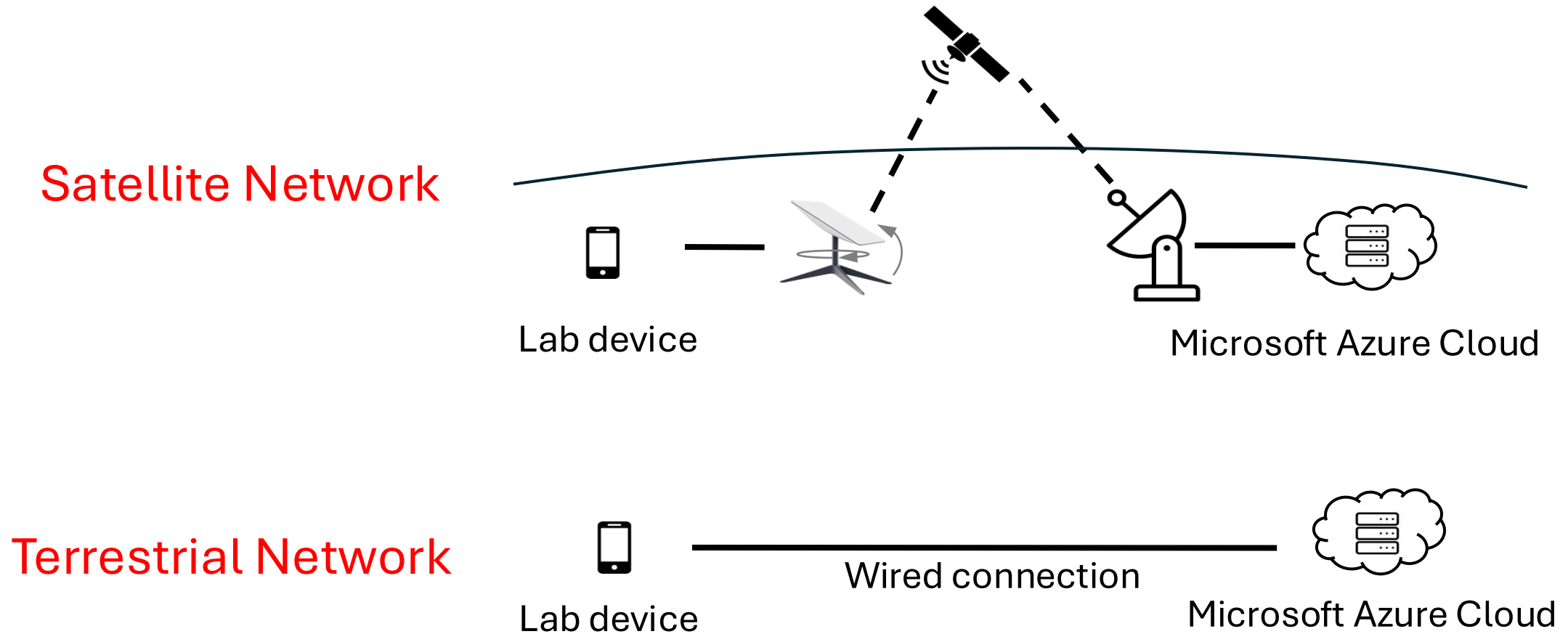


LEO Satellite Network

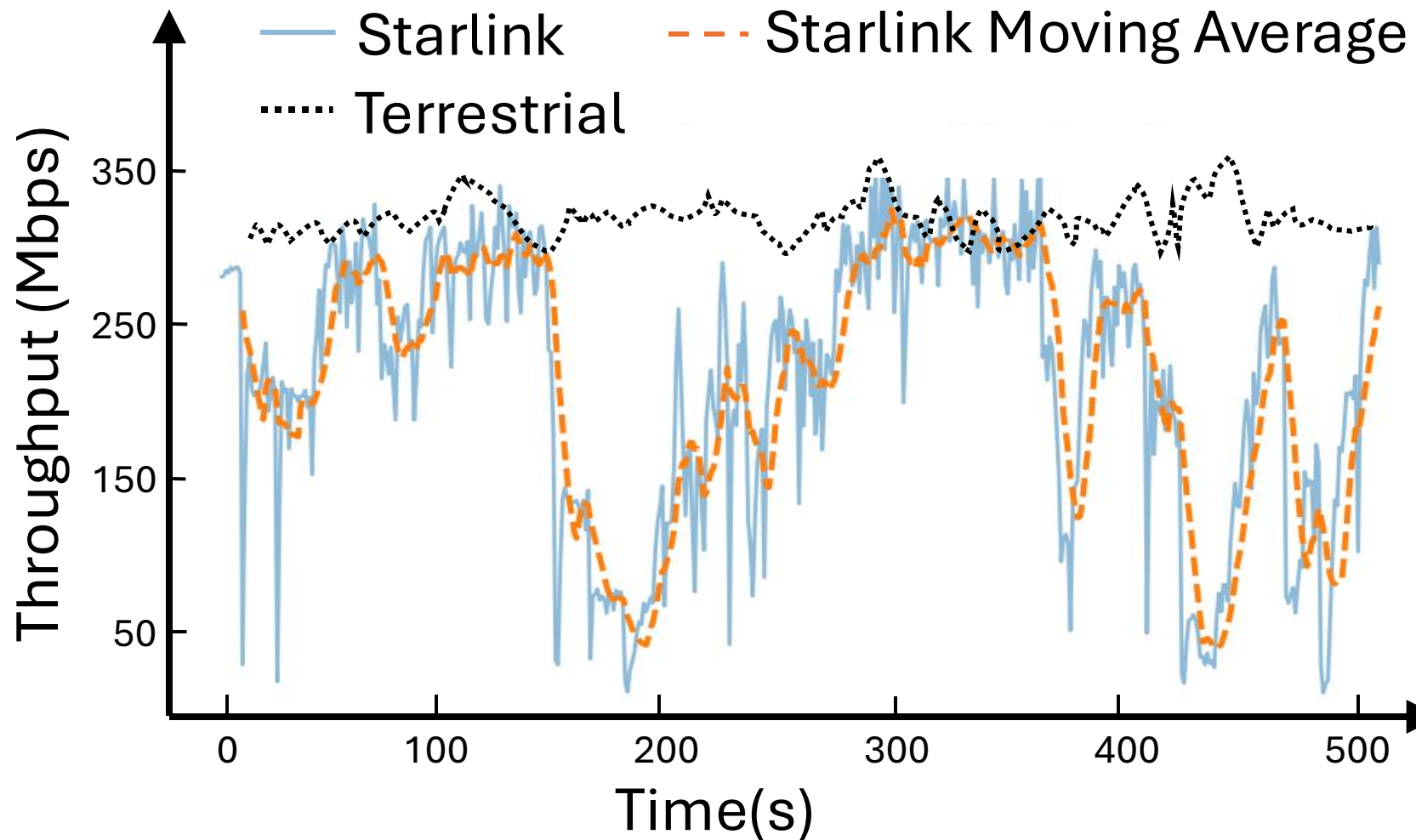
T3P...



Compare Starlink's throughput with terrestrial throughput



Challenge: Starlink throughput shows large volatility



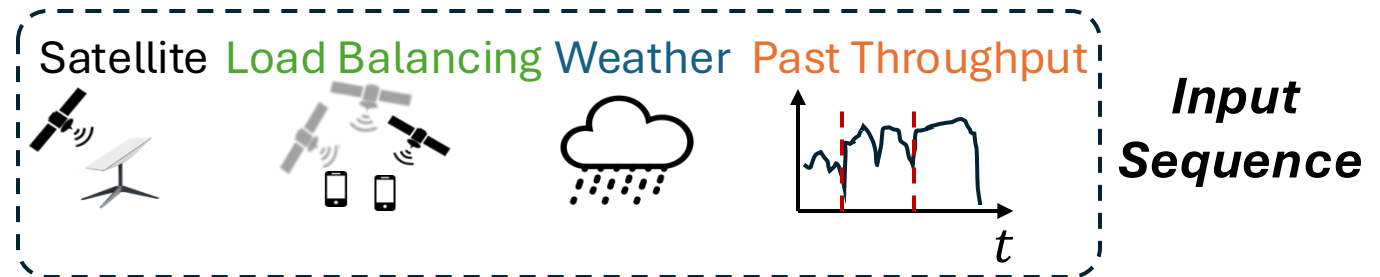
We propose StarNet

- A learning-based throughput prediction system
- Capture Starlink network unique characteristics
 - Overall trend of variety of factors
 - Periodical patterns
 - Attention mechanism



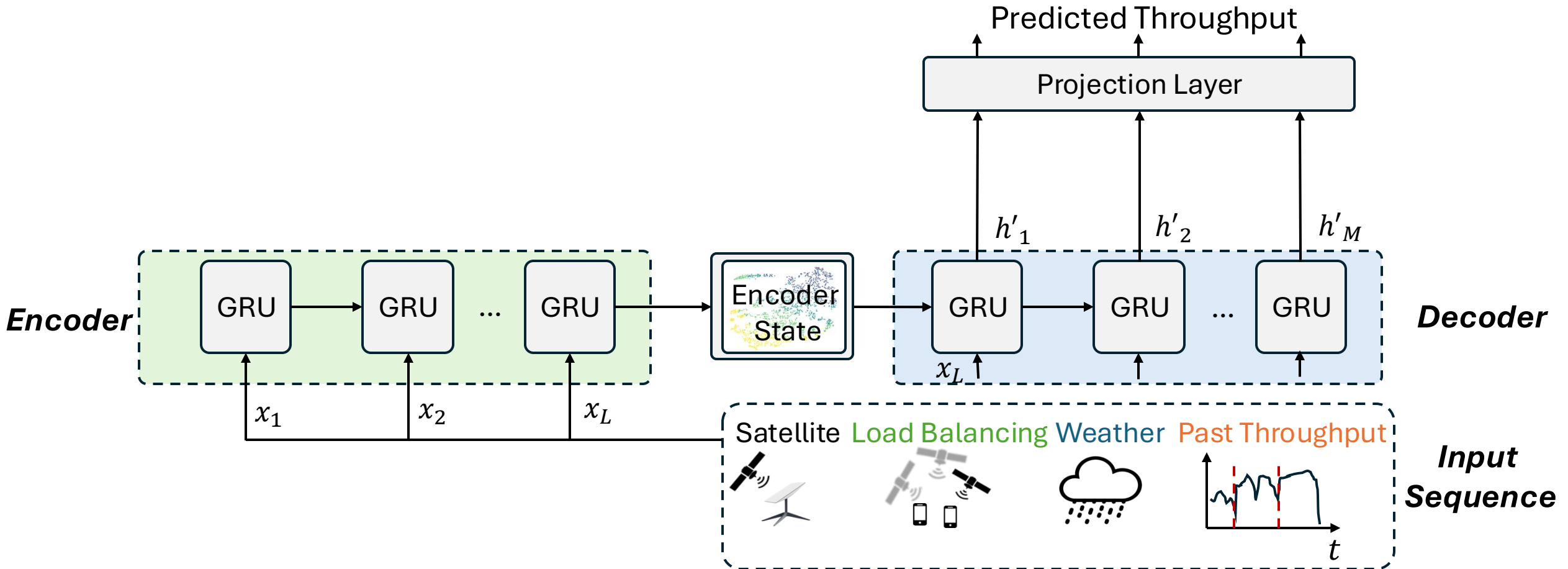
StarNet's Throughput Prediction System

1. Capture Varieties of Factors



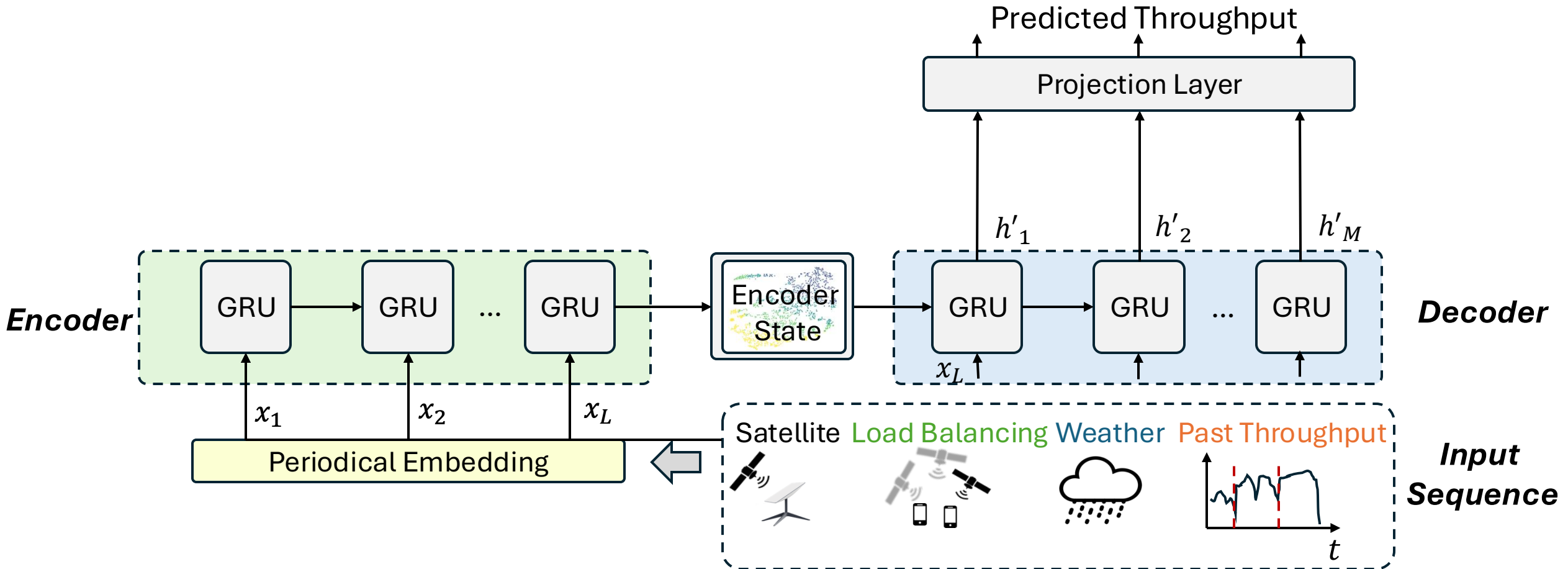
StarNet's Throughput Prediction System

1. Capture Varieties of Factors



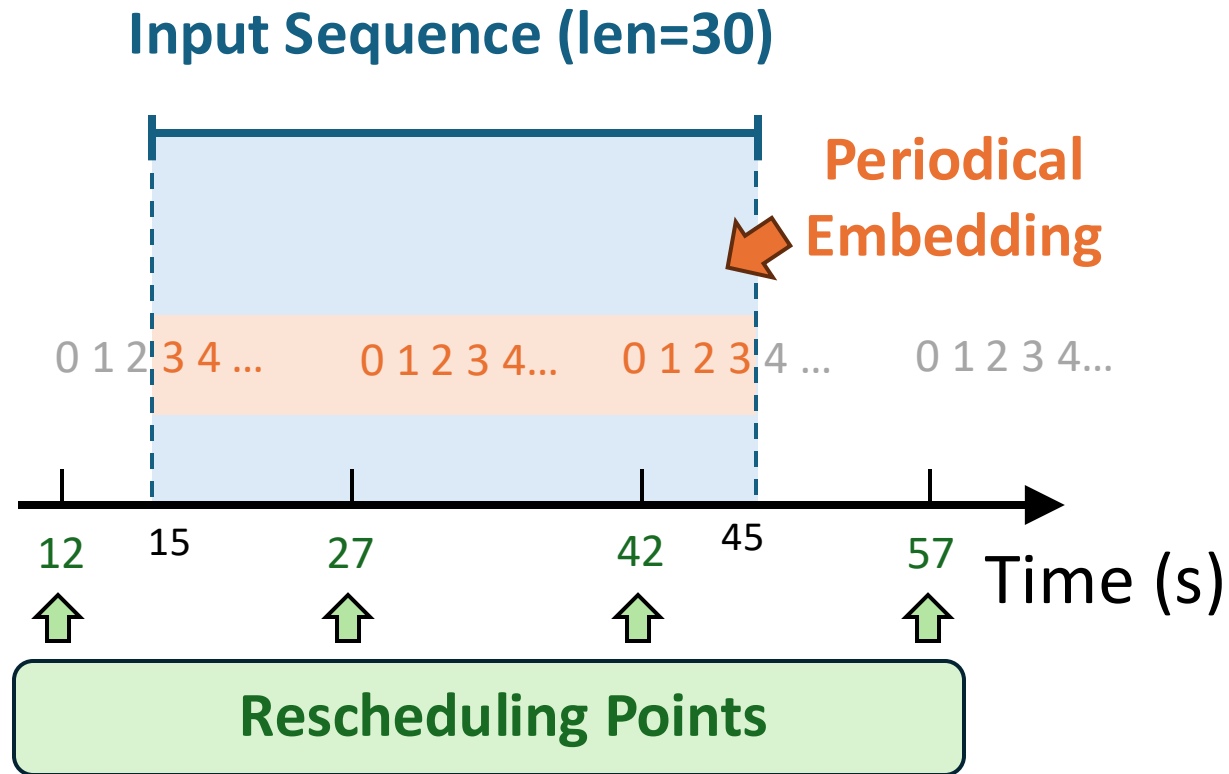
StarNet's Throughput Prediction System

2. Capture periodical patterns



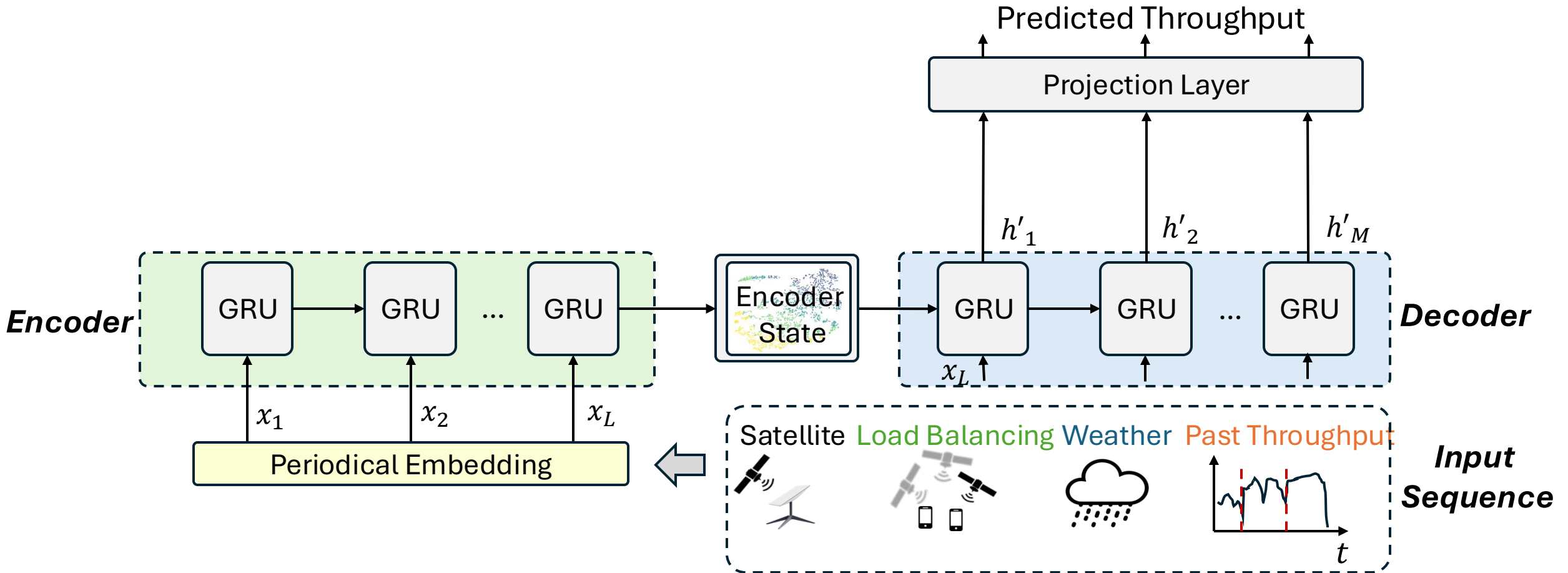
StarNet's Throughput Prediction System

Periodical Embedding



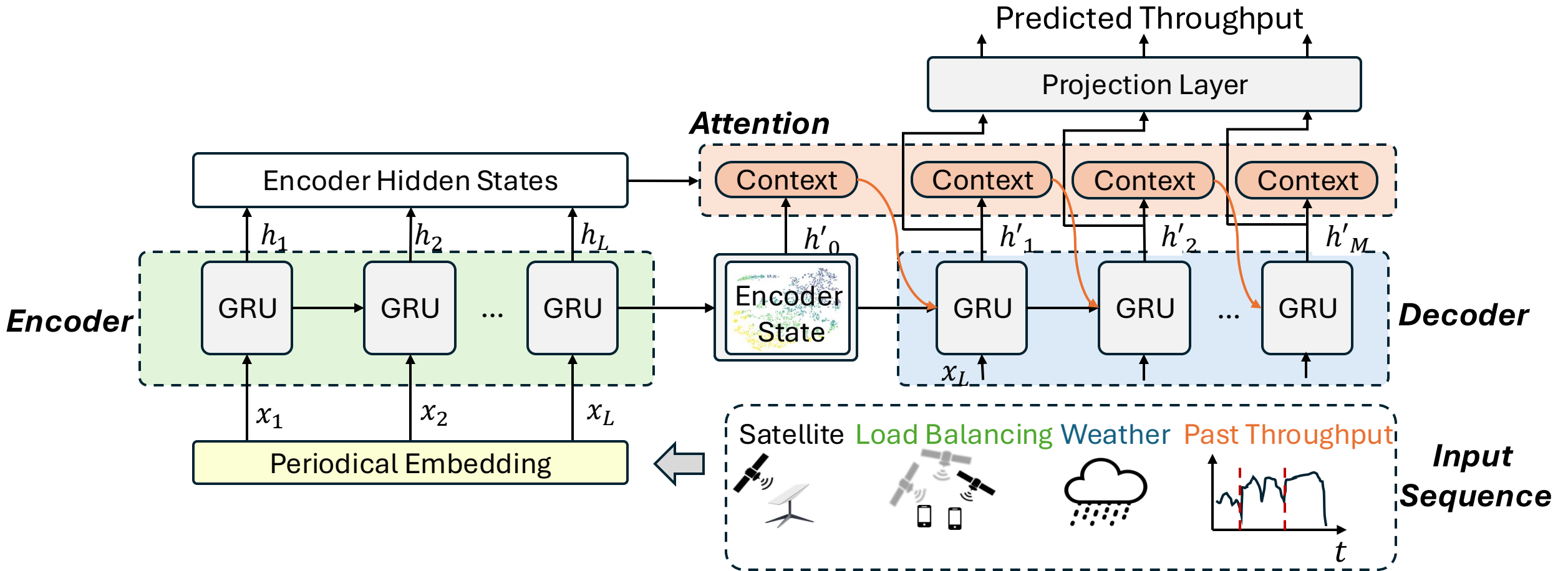
StarNet's Throughput Prediction System

3. Attention Mechanism



StarNet's Throughput Prediction System

3. Attention Mechanism

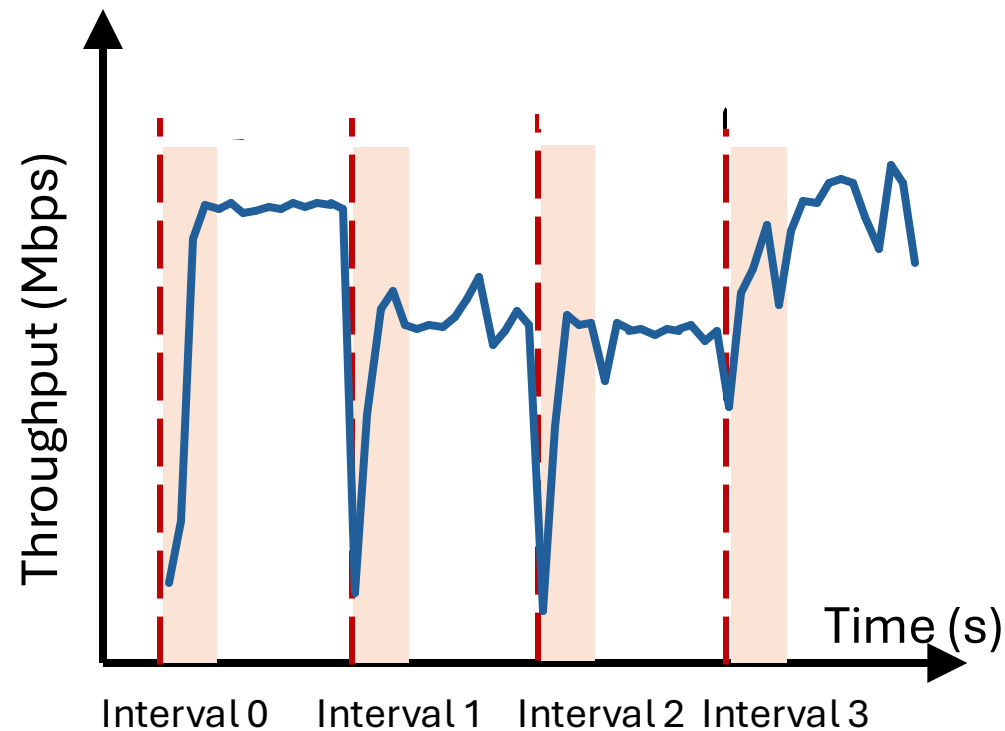


Evaluation

1. Model interpretability
2. Accuracy
3. Application

StarNet's Model Interpretability

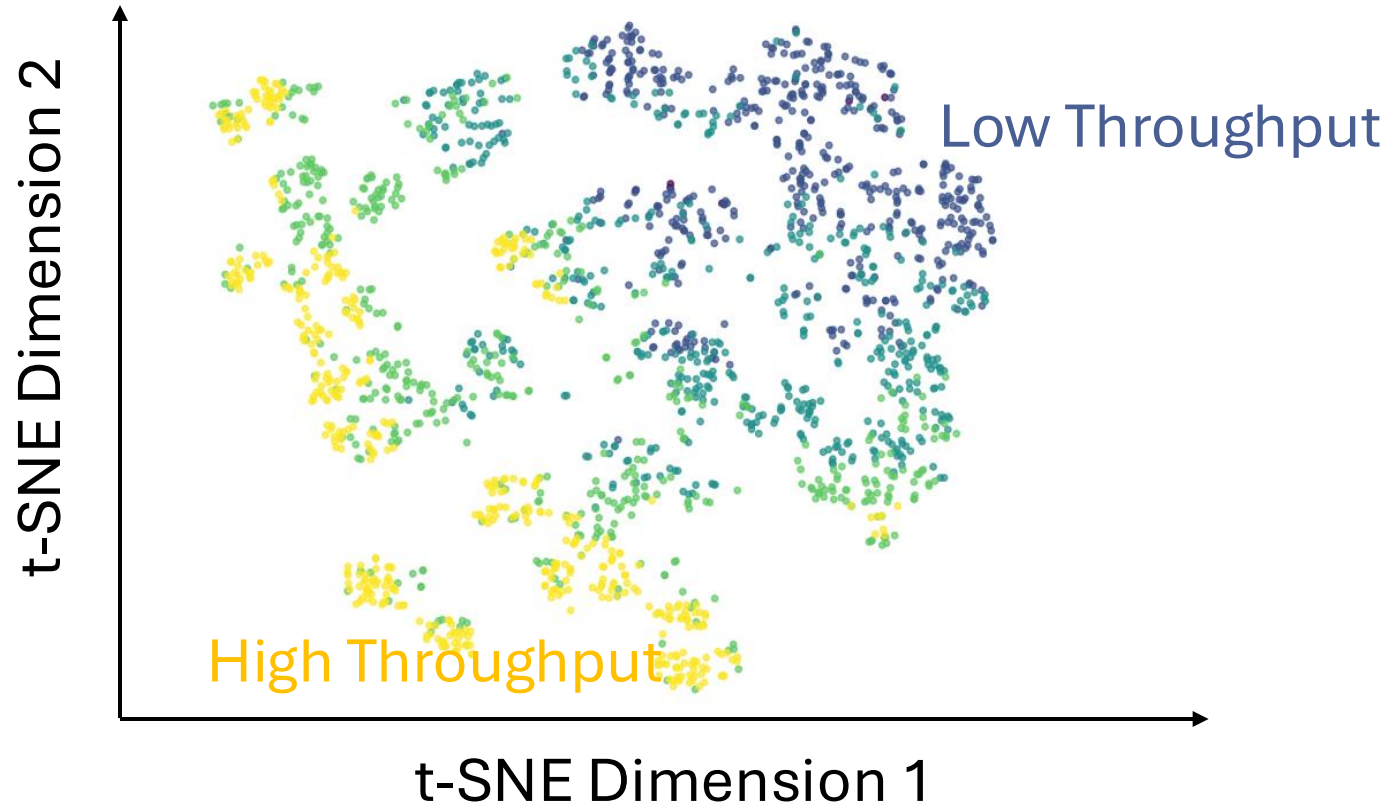
1. Attention Weights



StarNet focuses on the starting of the starlink interval

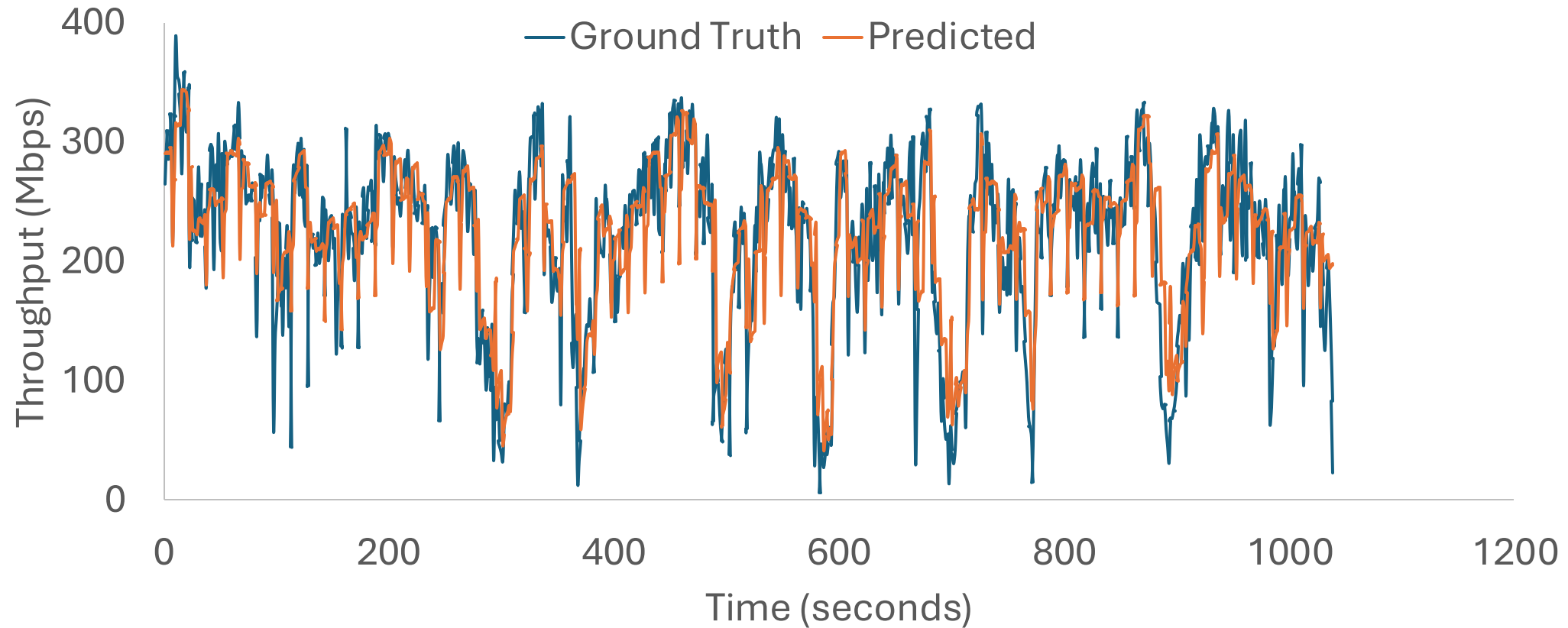
StarNet's Model Interpretability

2. Encoder State



The encoder state cluster is mapped to different throughput levels

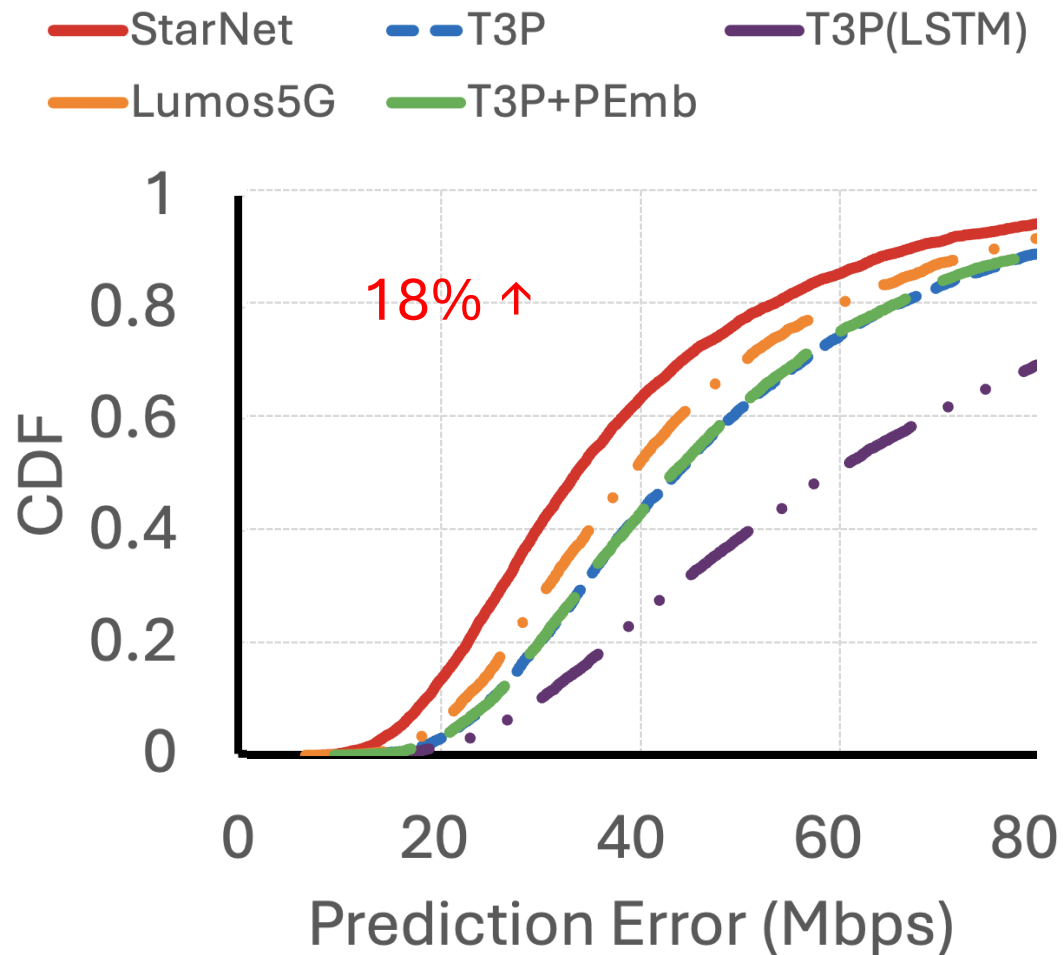
StarNet Enables Accurate Throughput Prediction



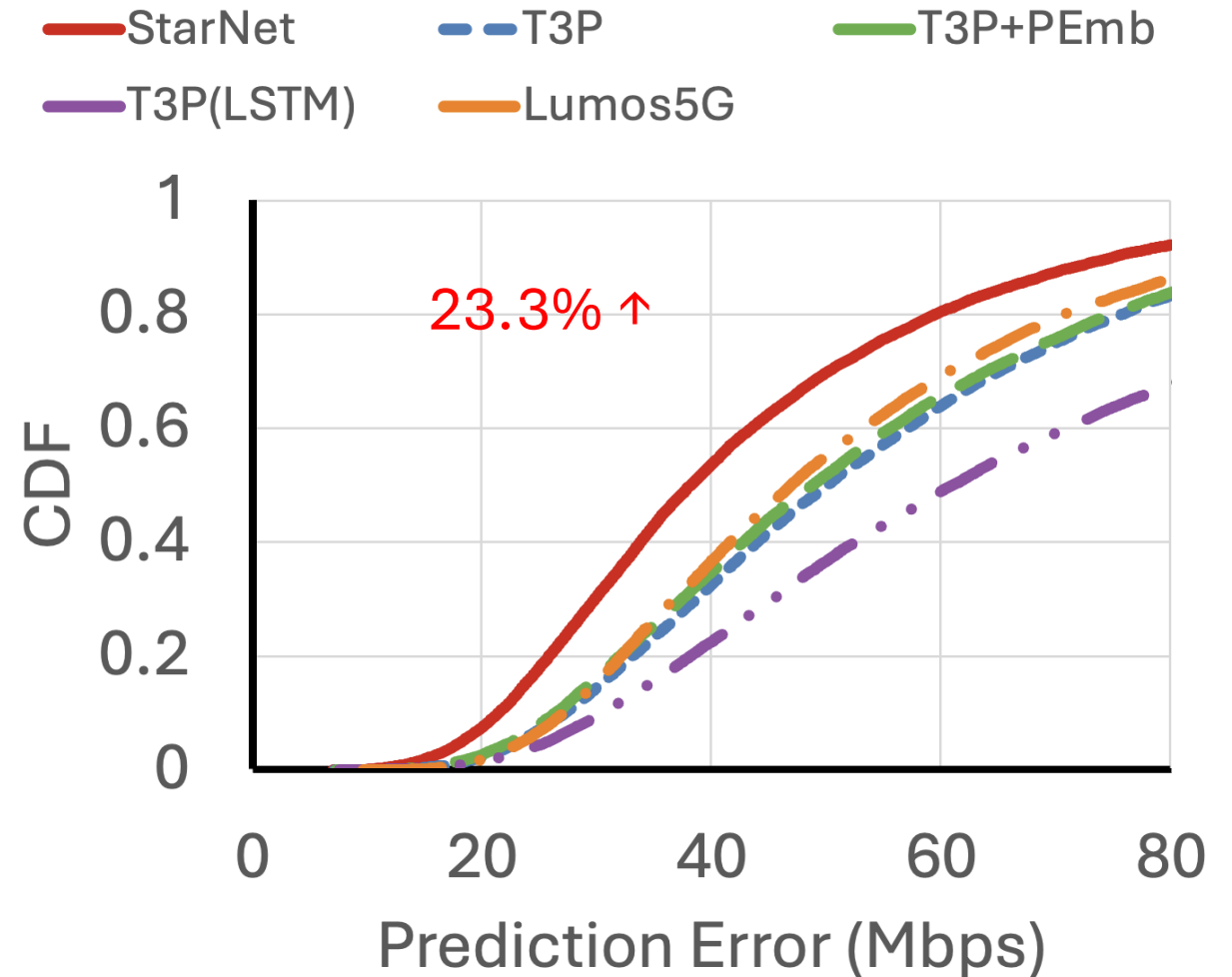
StarNet's Prediction MAPE: 16.1% < 20%

StarNet Prediction Accuracy Comparison

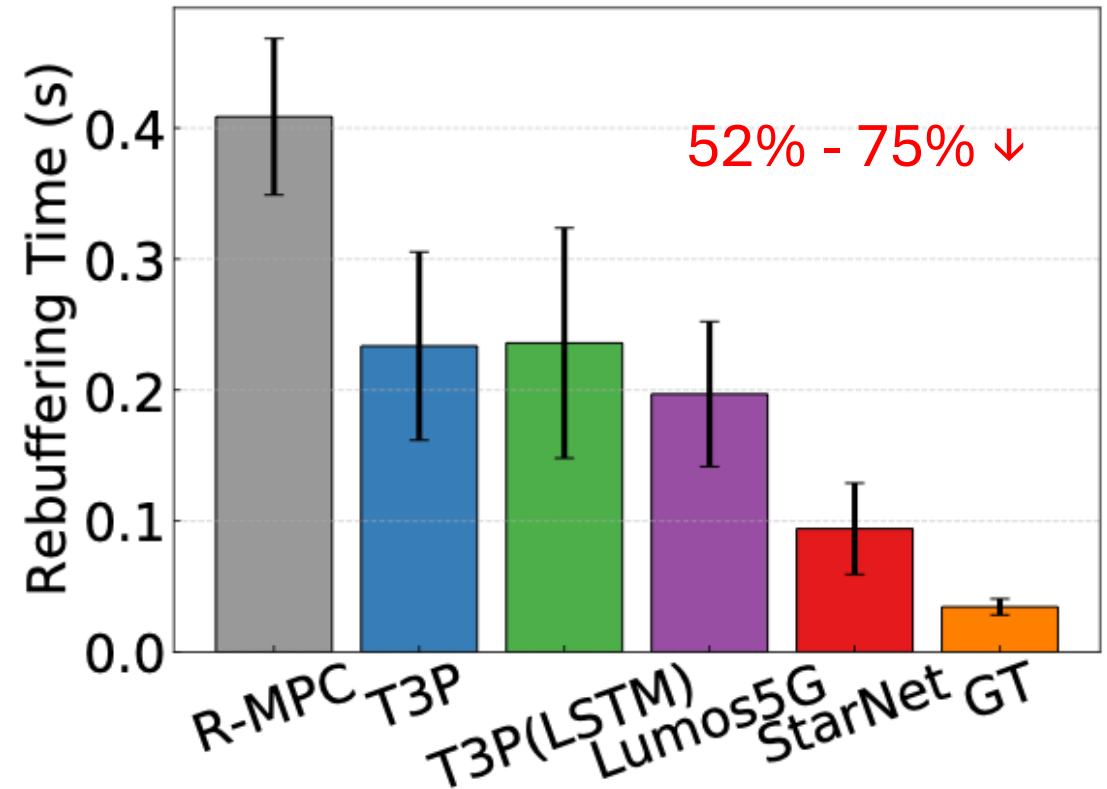
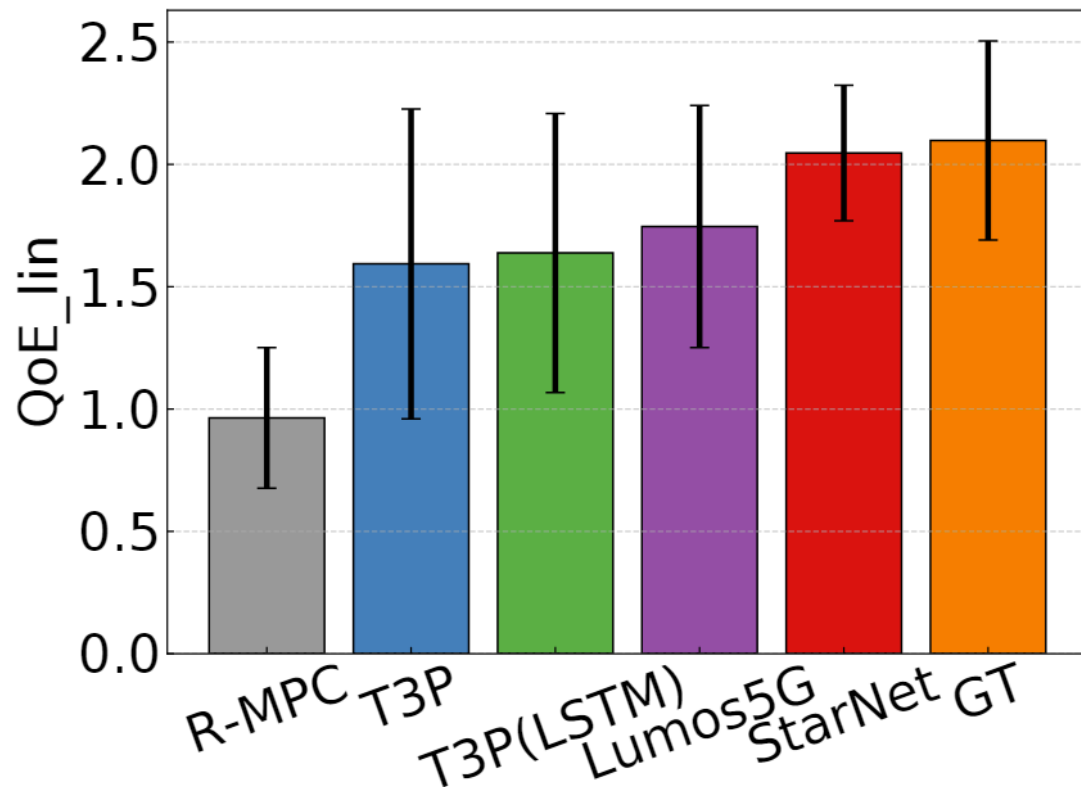
Compare with state-of-arts



Generalization Across Countries



StarNet Enables Better Video Streaming Experience



QoE Enabled by StarNet is only 3% lower than Optimal

Summary

- We conducted global Starlink **throughput measurement** with unique information such as the **serving satellite**
- We **identified key factors** that affect the Starlink's throughput
- We propose StarNet, a **throughput prediction system** that captures Starlink's network unique characteristics

Link to code:



Thank you!